

Stage 11 — standout analysis modules (each a standalone run_*.py, run in parallel as an independent Nextflow task)

Evolution & age	Regulation	Structure & engineering	Comparative & mobilization	Expression & benchmark
Subfamily phylogenetics run_phylo_analysis.py	Motif gain / turnover run_motif_gain.py	LTR structural annotation run_ltr_struct.py	Orthologous-insertion calling run_ortholog_insertion.py	Differential expression (clusters) te_expression.py
Repeat landscape / divergence run_divergence.py	Antisense / bidirectional promoter run_antisense_promoter.py	Protein structure (ColabFold) run_fold_prediction.py	Multi-assembly liftover run_multiassembly_liftover.py	Benchmark vs manual workflow run_benchmark.py
Automatic subfamily resolution run_subfamily.py	CTCF sites & TAD boundaries run_ctcf_tad.py	Allele-aware gRNA off-target run_grna_offtarget.py	3' transduction lineages run_transduction.py	
Consensus distance plot_consensus_distance.py	Epigenetic / regulatory overlay run_epigenetic_overlay.py			

All modules consume the clustered loci + consensus from the core pipeline and write their own figures and measured-value macros.